The Exponent Plane: Transfinite Asymptotic Residue pooling in Non-Commutative Geometries.

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10.5281/zenodo.23559868.
https://github.com/cymatic81/The-Exponent-Plane-Code-Base.git

The (Double Struck) 𝕏 Exponent plane is a number space where infinity has an axis. Thus allowing didactic operations, and others, of infinity as a variable on a plane, with asymptotic residue "pooling" for more accuracy. It is analogous to the algebraic function space of infinity in the exponent. It has similarities to the [C]Complex plane and the [No]Surreal plane. To be mathematically consistent a strict order of operations logic chain must exist, with non didactic operators incurring a cost. Logic QUALITY control through boundary rotations of infinity and operations MUST be strictly observed.

Axiom: Infinity to the power of infinity is Hyper.

Hyper exists only within the exponent of an exponent where an arbitrary division of infinity is made. The capital omega was chosen to be synchronous with sets containing infinity (is lower case better?), and Aleph which has a different index. The rules of hyper as a variable are the same rules as real numbers and infinity.

Hyper: H
Infinity: Ω

H:=Ω^Ω, Ω<H
H+Ω = H+Ω = H
H-Ω = H

Coming back out of the second exponent, what exponent operations now apply to infinity?

Linear Accumulation: Ω + Ω = 2Ω
Linear Retreat: 2Ω - Ω = Ω
Exact Self-Cancellation: Ω - Ω = 0
Scaled Cancellation: 3Ω - 2Ω = Ω
Multiplication (Dimensional Scaling): Ω . Ω = Ω^2
Division (Dimensional Reduction): Ω^2 / Ω = Ω
The Hyper-Division Wall: Because H = Ω^Ω, H / Ω = H

Some Division has no commutable portion.

Also of note, if these apply to Ω then they apply to: H
Hyper-Cancellation: H - H = 0 (possible use in proofs)
This cycle can be repeated, just as infinity can be repeatedly placed in an exponent. This is not used in the logic chain.

Axiom: Infinity is a logarithmic scalar and has a transfinite axis.
𝕏Exponent Plane Set (real {a, bΩ} real + infinity)

In support: Near the infinite smoothness of 1 or absolute 0. If some commutability is preserved it supports this New ruleset. Also, we should track the partially commuted/variant portion to identify the opportunity to exploit symmetries and return the partially commuted portion toward 0.5, when possible.

Ruleset: (logic chain)

1 Assumed Variant: (check notes for explanation)
Complete non-commutivity is assumed and given priority on resolution, unless commutability MUST remain. Rules only establish full data retention by jet bundles, partial data retention (asymptotic residue) is ignored, and is filled by full data retention MUSTS established by this rule set or is considered empty.

2 Quality Chain:
Quality MUST be tracked at all times. There is a Quality grading entering and exiting the Exponent plane (all boundary operators act here and give a quality grade to the function) and on Operations. Quality classes A, B, C and F and operation grades (+, -, 2-, ect) and (i) flags must be tracked and adhered to at all times.

3 Rotations of Infinity:
$R_{\Omega}(\theta) = \begin{bmatrix} \cos(\theta) & -\Omega \sin(\theta) \\ (1/\Omega) \sin(\theta) & \cos(\theta) \end{bmatrix}$
Different rotations of infinity on the transfinite axis determine the rotation matrix, and the quality of passage onto and off of the plane. The determinant is algebraically exactly 1 through ALL rotations of theta. Any approximation of symplectic area is explicitly not allowed.

4 Negative Infinity Axis:
$\Omega \cdot e^{(i \cdot \pi \cdot \Omega)} = \Omega \cdot (-1) = -\Omega$
Transfinite growth is reversed by a phase inversion in the negative axis.

5 Negative Real Axis:
As we are in exponent space (i) appears in the negative real axis and must be tracked as a pre-sub-script on the most recent Quality Chain Class (the one you created when you moved here).

6 Primary Operators:
Binary Multiplication: $x \cdot 2^n$
Binary Division: $x / 2^n$
Powers of 2 & Dyadic Roots: 2^n & $2^{k_root_of(x)}$
Addition and Subtraction: + and -
These operations are available at all times with no quality penalty to both axis on the [X]plane.

7 Secondary Operators: (excludes redundant repeats)
Non-Dyadic Multiplication: $x \cdot m$, where $m \neq 2^n$ [-1]
Non-Dyadic Division: x / m, where $m \neq 2^n$ [-1]
Non-Dyadic Powers & Non-Dyadic Roots: a^b, where exponent b is non-dyadic & $n_root_of(x)$, where index $n \neq 2^k$ [-1]
Additional Secondary operations are applied to both axes with regard to quality loss from orientations of infinity. They have a quality cost of -1. Of note 2 or more Primary operators can do some Secondary operations without penalty. Non-dydactic distribution shears function alignment at the real axis.

Definitions: Local Slope and Assessment Feild/Arc (no mathematical change)

* The “Eye” shaped Spindle can be assessed by a local slope slope = a / b(Ω). Stretching the local elipse arc to a straight line intercepting the dimensional axes makes a rhomboid.

* Concentric ABCF stratification layers of function distributions can be visually placed symmetrically above and below the local slope intercept, into an assessment field. Placed, in regards to their quality, bottom to top in the first quadrant, such that most aligned distributions (A) are nearer to origin, and the least aligned distributions (F) bend into their original shaped asymptote.

8 Quality Classes:
As a Function passes through the Xplane its trajectory is distributed to 4 quality grades A, B, C and F as it crosses the transfinite axis. Crossing the real axis is always a grade A crossing. Operators have a quality adjustment associated with them possibly changing the functions quality grade. These letters (with (i) flag) together with operators (with quality flags) determines the quality of the transit.
Functions can self promote within the Xfeild (depending on vector value) but it is rare and this ruleset is intended to catch all self demotions (iterating this rule set should catch all self promotions.) Each letter or operator can have a pre-sub-scipt containing the flags. Of note non-dydactic division and multiplication (distribution) shears the local slope alignment on the real axis.

Eg: [(-1)×3]B, A == C
Eg: [+]C, [/4]C == B (no shear and an additional symmetry)

A,A == remains A
B.B == remains B
A,B == remains B (defaults to lower grade)
C,C == upgrades to B (orthagonal infinities cancel)
A,C == remains C (can occasionally self promote from orthogonal infinities)
B,C == remains C (rarely self promotes from orthogonal infinities)
C,F == can upgrade to C (from orthagonal infinities)
F,* == remains F (F,* can benefit in data accuracy, and occasionally can self promote from orthogonal infinities)

9 Collapse to Real:
As two vertical lines travel in from infinity equally on the real axis, they stop at the first intersection of negative and positive infinities for resolution (the first rhomboid they touch.) All intersects must be assessed regardless of their quality. When there are no intercepts then assess the local slopes for asymtopes. The (i) flag decides the sign in the real axis. An A/B might be forced out on a C exit, or can chose the C exit to with a different value of infinity (max one infinity!) Leaving the Xplane to real ceases the need to track quality.
Classes A+B
Can choose, and take the real axis exit with any valid amount (-1, 0 or 1) in the infinity axis (in the exponent)
Classes C
Will have one answer.
Classes F
Are still asymptotic, and are identified by the collapse to their local slope intercept, taking careful note of higher order functions that may appear to be asymptotic. Often the new asymptote on the

Xplane is more accurate, benefitting from “pooling” the best partially commuted portion, and can produce better approximations.

Addendum: Tertiary Operators (All 𝕏 Plane boundary operations are tertiary class and are subject to the quality chain)
(not in the logic chain, not tested)
Inbound Operations: An Inbound Operation must be initialized by a Boundary Entry Operator and terminated by a Boundary Exit Operator (inbound can be boundary)
Calculus and other approximations

Set Extraction applying a function to a set/matrix

Boundary Operations: Boundary Operators must open and close the scope of an Inbound Operation, and quality must be tracked for any interaction with the Xplane.
Sin0/180 cos 90 bypass: Alternate proof, add 1 Ω axis to a Euclidian space.

[O]Ontologicalspace: 3 dimensional axes; real, imaginary, infinity. (No (i) flag needed)
It just is, and appears to commute.

The Exterior Ring Baffle Rule (destroy data rule) Manifold Boundary Compactification. Enclose the infinite perimeter with a swappable boundary baffle, using asymptotic limits to capture wayward tangents and govern asymptotic quality. A “proof plane” fit with infinite tightness on the transfinite axis. Ensure the siplectic area remains the same after fitting the Baffle. If the area is not alegraically EXACTLY = 1 the manifold does not fit! Get the right manifold for your geometry today! (all geometries not included, any use voids warranty.)

*Poincaré Disk Model: Enforces 90-degree orthogonal asymptotic residue matching baffle.
(Seems to improve asymptotic residue when mapping transinfinite axis to real axis. Sudo-Solid Tension Field)
*Logarithmic Spiral Surface: Tracks rotational momentum and phase shifts.
*Radial Compactification Disk: Preserves isotropic scale and smooth macroscopic boundary flow.

Hyperplane (Hyperspace): A geometry disregarding “location” whilst arbitrarily connecting to one or more overlaid transinfinite Ω axes with a logarithmic scalar of hyper. (The wormhole)

Notes:

Data replacement: When partially commuted data is replaced, if the rotation that caused partial commutability is tracked and compared to the rotation of of the replaced data, additional accuracy could be achieved, beyond what this ruleset provides. Applying this ruleset to “Hyper” could allow for even more accuracy than data replacement above, where rotations allow. This, and the data replacement improvements process would be multiple times more demanding for an ever so tiny improvement, with a decreasing opportunity to allow it. Rotations could also be tracked on ALL inbound operators starting as a 2nd level iteration for an even more minute benefit. (Digitally iterate this!) There should be some set analysis to provide the algebra of this process iterated, however that is well beyond the scope of this project.

The ruleset is complete When rule 1 becomes: (loosening the restriction into the logic rules.)

1 Assumed Variant:

Complete non-commutivity is assumed. Except when complete data transfer is available and in turn given priority (and is often a confirmation.)
Partial data transfer is allowed but assumed Non-commutive, unless quality is attached. Partially commuted Data replacement is determined by 1) allowed to replace, 2) is better quality.

Documentation of process

The Exponent Plane just is. I have been mentally standing in the 1st quadrant of the plane doing maths problems for longer than I knew it existed. It has a lot of similarities to the exponent place of BigO log2 n, where coders visit frequently.
This document is my best attempt at formalising the process I was using and I believe it is a very useful tool for mathematics. I wasn't fully using the phase inversion! and I'm still not sure of it's full benefit.

I intend to use this document as a specification for a new implementation of infinity in code. I apologise for the nomenclature as I do not have a university education. The closest University is a two hour drive I have not been able to get scholarly input thus far (mathematical science is not student fees.) This is a large project for me and it is nigh impossible that I made no mistakes, despite every effort to be precise. This work needs to be ruthlessly scrutinised for it to be of benefit. Please give feedback with improvements and criticism disregarding my feelings as that will be more useful. The race has begun. Please beat me to the finish line!

The axiom about hyper is to show how the exponent rules apply to infinity. It is essential to the Xplane, derived from algebra. To allow clean set extraction notation, I chose capital omega as the representation for infinity to avoid confusion, lower case would be the 2nd best option IMO, but Aleph works too despite index (TCH it don't matter.) The rules derived from algebra that apply to infinity are listed, taking note of the limited division operations, didactic division is established with the dimensional scaling aspect of infinity squared. The process can be repeated starting with an arbitrary division of hyper, enabling all the same rules to apply. The Hyper cancellation helps show what is happening in the exponent.

The process in getting to this point.

The Ruleset is formatted so that someone can follow it step by step to calculate any advantage provided by asymptotic residue pooling. It is also a logic chain that must be followed to accurately bring the algebra from the exponent to the geometry of the exponent plane. Order of operations from the exponent make for some unusual conditions on the Xplane resulting in a more limited set of operations on the field by comparison to a Euclidian Plane.

1. Assumed Variant:
Rule 1 from this document should NOT be used, it will be less accurate than current methods not tracking asymptotic pooling, or at best be the same. Rule 1 from Notes should be used instead. I have chosen to leave Rule 1 in this document as a record of the process I used to develop this rule set. Using this initial rule 1 may help you copy my process to improve the Xplane as you see fit.

Complete non-commutability, whilst being somewhat meaningless, is also nessecary so that the Xplane remains analogous to the algebra it is derived from. By data or jet bundles I mean the geometric information, the tensor of the axis where commutability is preserved over the transfinite axis, and partial data is the tensor that gets "blurred" by it's association to infinity. The best example

I can give for this is when taking a tangent between two geometries, one part is preserved, the other is discarded but is partially preserved.

The alternate rule 1 from notes should be used, and after sufficient scrutiny of the complete ruleset it should be promoted to it's rightful position. As this rule is worded now, it is still allowing "slipperryness" to the data accumulation in the pool. In the operational space of a function, however the data transfer happens as you calculate, and the most recent partially commuted data is used. Data from asymptopic residue of functions with A,B qualities share equal precision and are allowed to replace any data. Class C functions have lower quality asymptotic data and can only replace F data which is all asymptopic inherently.

2 Quality Chain:
In order to remain analogous to the exponent, we implement a quality chain. As a function passes through the Xplane there is a distribution of the general form. The local form of the curve is an ellipse. We declare 4 divisions of this distribution derived from the rotations of infinity, or the full or partial jet bundle data. A, B and C remain tight to the function's general form, whilst F has a asymptotic relation to the real axis (or a higher order) and never intersects it. Perfoming non-didactic operations on the Xplane incurs an additional quality cost above the inherent boundary rotations, how ever this can be done without causing all local functions to miss the real axis, leaving you with a valid answer.

3 Rotations of Infinity:

$$R_{\Omega}(\theta) = \begin{bmatrix} \cos(\theta) & -\Omega \sin(\theta) \\ (1/\Omega) \sin(\theta) & \cos(\theta) \end{bmatrix}$$

This is where the mechanics of the asymptotic residue occur. As a function crosses the transfinite axis it always has a determinate of 1, no matter what the vector value is at the boundary. The Simplectic Area must be precisely evaluated, absolutely no floating points in code!
This part of the process is in need of close examination, I have done my best here, but… ouch.

4 Negative Infinity Axis:

$$\Omega \cdot e^{(i \cdot \pi \cdot \Omega)} = \Omega \cdot (-1) = -\Omega$$

I really don't know what to say about this, it just is. I'm also well aware that this is insufficient. Ω is a valid co-ordinate and (i)imaginary numbers are inherent to the Xplane so the phase inversion applies, despite being an illegitimate algebraic operation. (Functional analysists please help) The best functional analogy I can give is: bringing a limit down, whilst having the inverse limit go up, until they meet at the answer. The involution at the boundary results in (i) acting as a phase operator, but not a scaling multiplier in relationship to infinity.

5 Negative Real Axis:
Imaginary Numbers are inherent to all algebraic functions. The (i) flagging method is redundant, and this space is inherently a 3 dimensional space.
However, to keep complexity to a minimum I have chosen to track it as a flag during the development of the Exponent Plane.

6 Primary Operators:
These operators are derived from the first axiom, and the rules it brings with it.
These are inbound operations of the Exponent plane. Of note there is only dydactic distribution within the primary operations. The tiers of operations introduced were chosen so as not to be confused with other operation nomenclature. These operations are identical to the BigO log2 n space. It is how mathematicians cancel infinities at will in the algebraic space. It is the logic gate space of deterministic binary Markov chains.

7 Secondary Operators:
Whilst these operations are not inherent to the Xplane they are available at cost. This quality cost comes with the action of shearing function alignment on the real axis. The dyadic valuation ring of the Exponent plane causes p-adic grid shearing as the transfinite axis only allows discrete values. The inherent base 2 confirms the valuation of the cost. Of note two or more primary operations can be used instead of a secondary operation, just as binary Markov chain collapse can be done in different ways to achieve the same outcome.

Definitions:

————graphs needed————

I really hate arguing with the word processor.

8 Quality Classes:
Here is where I outline the tracking notation. 1 Scalar and 5 identities of binary states track quality. Consisting of 2 entry rotations, 2 exit rotations and an (i) flag. One positive only scalar for shear is needed. This is the complete table needed to implement this ruleset. The rotations are orthogonal or tangential and when crossing the real axis are both orthogonal. The cost of secondary operators increases the shear count. The ruleset it self is simplified as much as possible whilst reming explanatory.

9 Collapse to Real
All of the tracking to this point decides which of the many local functions have valid intercepts. The largest intercept is correct due to the smaller intercepts being an internal harmonic of the logarithmic environment. In Transfinite geometry the outer most function is at the boundary of stability before shearing and is considered correct, there can be more than one. Functions can exit the exponent plane with 1, 0 or -1 infinities as valid answers depending on need.

To document the addendum:
The addendum is beyond the scope of this project. It signifies where the exponent plane fits in relation to geometry in general. Identifying boundary operations and how they interact with the quality chain. A lot of what is included there is what I noticed as I was undertaking this project.

To document the notes:
Rule 1 has been adequately covered. The data replacement identifies how I believe this should be progressed. This process needs to be digital and when that is tried and tested, it should be iterated, easier typed than done. This should then be able to be applied to higher dimensional spaces too.

Code Base
https://github.com/cymatic81/The-Exponent-Plane-Code-Base.git

The Following is the C code implemented from this document. It appears to be working but is NOT completely edge tested. Take any evaluation it may give with CAUTION!

```
#include <stdint.h>

#define XPLANE_UNINITIALIZED 0xDEADBEEFDEADBEEFULL
#define XPLANE_INF 0x7FFFFFFFFFFFFFFFULL
#define XPLANE_OMIT_PARAM 0xFFFFFFFFFFFFFFFFULL
#define XPLANE_MASK_QUALITY_F 0x8000000000000000ULL

/* Rule 9 Structural Exit Vectors */
#define XPLANE_FLAG_EXIT_REAL 0x0000000000000100ULL
#define XPLANE_FLAG_EXIT_POS 0x0000000000000200ULL
```

```
#define XPLANE_FLAG_EXIT_NEG 0x0000000000000400ULL

/* Exact RDX Hard Register Masks */
#define XPLANE_MASK_SHEAR_COUNT 0x0000000FULL
#define XPLANE_MASK_I_FLAG_COUNT 0x00007FE0ULL
#define XPLANE_MASK_MULTI_SHEAR 0x07FF8000ULL
#define XPLANE_MASK_EXIT_MULTIPLICITY 0x18000000ULL
#define XPLANE_MASK_ROT_OUT 0x60000000ULL
#define XPLANE_MASK_ROT_IN 0xC0000000ULL

#define XPLANE_EXIT_INF_NEG 0x08000000ULL
#define XPLANE_EXIT_INF_POS 0x10000000ULL

typedef enum {
XPLANE_RUN_DEFAULT = 0x00,
XPLANE_OPT_ACCURACY = 0x02, // 128bit-65bit accuracy toggle.
XPLANE_NO_POOLING = 0x04, // Xplane Bypass.
XPLANE_DISCARD_RESIDUE = 0x08, // Abstractly: take the tangent.
XPLANE_OPT_EXIT_INF_NEG = 0x10, // -1 Infinity Exit
XPLANE_OPT_EXIT_INF_POS = 0x20 // +1 Infinity Exit
} xplane_opt_t;

typedef struct {
uint64_t Real;
uint64_t ShearRollover;
uint64_t Imaginary;
uint64_t Flags;
} RealIntercept;

void mathlib_xplane_evaluate(RealIntercept* node, xplane_opt_t options, ...) {
if (!node) return;

/* HARD REGISTER GATE: Evaluate initial Rollover Shear limits (Bits 0-3) first */
uint64_t rollover_shear = node->Flags & XPLANE_MASK_SHEAR_COUNT;
if (rollover_shear >= 9) {
if (!(options & XPLANE_OPT_ACCURACY)) {
node->Real = XPLANE_UNINITIALIZED;
node->Flags = XPLANE_MASK_QUALITY_F;
return;}}

/* 1. VA_ARG ONBOARDING AND REGISTER FLAGS CYCLING */
uint64_t t1 = 0ULL;
uint64_t t2 = 0ULL;

va_list args;
va_start(args, options);
uint64_t custom_t2 = va_arg(args, uint64_t);
uint64_t flag_arg;
while ((flag_arg = va_arg(args, uint64_t)) != XPLANE_OMIT_PARAM) {
node->Flags |= flag_arg; }
va_end(args);

/* 2. DYNAMIC OPERATIONAL OVERRIDE: Sync option enums to RDX register structures */
```

```
if (options & XPLANE_OPT_EXIT_INF_NEG) node->Flags |= XPLANE_EXIT_INF_NEG;
if (options & XPLANE_OPT_EXIT_INF_POS) node->Flags |= XPLANE_EXIT_INF_POS;

/* 3. RULE 7: EVALUATE TRANSFINITE SHEAR FROM OPERATIONS AFTER INITIAL ROLLOVER SAFETY */
uint64_t operations_shear = (node->Flags & XPLANE_MASK_MULTI_SHEAR) >> 15;
uint64_t total_shear_profile = rollover_shear + operations_shear;
if (total_shear_profile >= 9) {
if (!(options & XPLANE_OPT_ACCURACY)) {
node->Real = XPLANE_UNINITIALIZED;
node->Flags = XPLANE_MASK_QUALITY_F;
return;}}

if (custom_t2 != XPLANE_OMIT_PARAM) {
t1 = node->Real;
t2 = custom_t2;
} else {
t1 = node->Real & 0x5555555555555555ULL;
t2 = node->Real & 0xAAAAAAAAAAAAAAAAULL;}

/* 4. OPTIONAL MODIFIER ROUTING: DISCARD RESIDUE TANGENT MODE */
if (options & XPLANE_DISCARD_RESIDUE) {
node->Real = node->Imaginary;
node->Flags &= ~XPLANE_MASK_QUALITY_F;
return; }

/* 5. RULE 4 PHASE INVERSION (Extract (i) count from Bits 5-14) */
uint64_t winding = (node->Flags & XPLANE_MASK_I_FLAG_COUNT) >> 5;
if (winding & 1) {
t1 = ~t1;
t2 = ~t2;}

/* 6. UNROLLED MACHINE CODE BINARY SEARCH PASS */
uint64_t low = 0ULL, high = 0xFFFFFFFFFFFFFFFFULL, mid = 0ULL;
for (int i = 0; i < 64; i++) {
mid = low + ((high - low) >> 1);
if (mid == t1) { t1 = mid; break; }
if (mid < t1) { low = mid + 1ULL; }
else { if (mid == 0ULL) break; high = mid - 1ULL; }}
if (t1 != mid) t1 = low;
/* 7. RULE 9 COLLAPSE & MULTIPLICITY EVALUATION */
if (t1 == 1ULL) {
node->Flags |= XPLANE_FLAG_EXIT_POS;
if (node->Flags & XPLANE_EXIT_INF_POS) {
node->Real = XPLANE_INF;
} else {
node->Real = t1;}
} else if (t1 == (~0ULL)) {
node->Flags |= XPLANE_FLAG_EXIT_NEG;
if (node->Flags & XPLANE_EXIT_INF_NEG) {
node->Real = 0ULL;
} else {
node->Real = t1;}
} else {
```

```
node->Flags |= XPLANE_FLAG_EXIT_REAL;
node->Real = (options & XPLANE_NO_POOLING) ? 0ULL : (t1 | t2);}}
```

Code Base Explaination

The Following is the best explaination I can give of whats happening here. Do you remember the time you learnt about binary markov chains? And how they interact with hardware? Well do I have a story for you…

The function uses two values, for each the positive and negative values of infinity. They sweep into each other and meet on the real axis at the evaluation in a bitwise precise logarithmic p-adic evaluation ring. A binary search across 64 bits is done to find the point at which they meet. The "flags" aquired or set on function call will determine the evaluations valid exit point and quality. Logarithmic shear, and (i) is bit tracked directly to the register or on the stack when flagged. All these conditions are computed during the binary search in 1 tick look up as 64 bits is being searched.

Furthermore, the code is only in development and there is additional enum available for added functionality, for now you will have to wrap that functionality around this function. There will be an optimal arrangement for this enum table that will be discovered in time. Once that is found this whole function itself can be further wrapped up into a markov table for even faster execution.

Euclid has multiplied the suns in the galaxy available to humainty for compute.